



A numerical study of intra-catalyst diffusion limitations in small-scale ammonia plate reformer

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ABSTRACT

Ammonia is an efficient hydrogen carrier that can be easily converted into H₂-rich gas. In this study, a numerical investigation of the ammonia decomposition process within a small-scale plate reformer is presented to understand the effect of catalyst layer thickness on intra-catalyst diffusion limitations. Simulations are conducted using a RANS approach coupled with chemical kinetics, diffusion, and porous medium models. The different catalyst layer thicknesses ranging from 100 μm to 2000 μm are examined over a temperature range of 400 to 700 °C. The numerical results are compared with experimental data obtained for unmodified Ni-Al₂O₃ catalyst, revealing a strong correlation with an average deviation of less than 4%. The results reveal the significant influence of catalyst layer thickness, particularly up to 1000 μm, on the ammonia decomposition process due to intra-layer diffusion limitations. Beyond 1000 μm in thickness at temperatures between 500–600 °C, further increases in thickness showed minimal impact on ammonia decomposition and hydrogen yield. The effect of temperature on diffusion limitations is minimal. Additionally, at temperatures exceeding 650 °C, close-to-stoichiometric product gas composition was observed for low Reynolds numbers, even with a catalyst layer thickness of 100 μm.

1. Introduction

The current technologies for utilizing fossil fuels demonstrate high thermal and economic efficiency in various applications, e.g. power generation, internal combustion engines, etc. [1,2]. However, the global shift toward a carbon-neutral economy is driving the focus towards the adoption of carbon-free fuels that produce zero carbon dioxide emissions upon combustion [3–5]. Among these options, hydrogen has garnered significant interest as a potential alternative to traditional fossil fuels [6,7].

While hydrogen shows promise as a clean energy carrier, its widespread utilization faces challenges related to storage and transportation due to its low energy density (in gaseous form) and high flammability [8,9]. These limitations have prompted research into alternative solutions, with ammonia emerging as an excellent hydrogen carrier [10,11]. Ammonia as a hydrogen carrier offers several advantages, including higher energy density (comparing to other hydrogen carriers) and the ability to be stored and transported using existing infrastructure [12,13].

The conversion of ammonia to hydrogen-rich gas through decomposition provides a viable pathway for efficient and clean hydrogen

production [14,15]. This process is particularly attractive for on-board hydrogen production, where hydrogen can be generated prior to utilization [16,17]. Such an approach addresses the storage and transportation challenges associated with gaseous hydrogen while maintaining flexibility in meeting energy demands [18,19].

The ammonia decomposition process is characterized by its endothermic nature, requiring an external energy supply to convert ammonia into hydrogen or hydrogen-rich gas [14]. Renewable electricity generated from wind or solar energy emerges as a promising energy source for facilitating the production of hydrogen from ammonia [20, 21]. In addition, direct solar radiation can be considered as an energy source for ammonia decomposition. Bingzheng Wang et al. proposed a method for hydrogen generation using solar energy by combining a solar concentrator with a catalytic membrane reactor [22]. They achieved 99% ammonia conversion at 200 °C and 10 kPa pressure, demonstrating high efficiency at low temperatures. This study highlights the potential of integrating ammonia decomposition for hydrogen generation with solar thermal technologies.

Therefore, the utilization of renewable ammonia with renewable energy sources for its decomposition provides a carbon-free solution

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for hydrogen production. In this paper, we focus on heat as the primary energy source for the ammonia decomposition process. Ammonia reformers are commercially available in different configurations, including tubular, foam, and plate-type designs, each suitable for either large- or small-scale production. The reaction chamber can be activated using either packed catalyst particles or wall-coated catalyst layers. While highly active precious metal catalysts like Ruthenium can achieve complete ammonia conversion at lower temperatures ($\sim 450^\circ\text{C}$) [23], nickel-based catalysts are a more economically viable alternative, which is the focus of the present numerical study [14].

Itoh et al. proposed a novel tube-wall catalytic membrane reactor (TWMR) integrating two key innovations: a tube-wall catalyst configuration for enhanced heat transfer and a thin ($2\ \mu\text{m}$) palladium membrane for selective hydrogen removal [24]. The synergistic design achieves high performance, demonstrating near-complete (100%) ammonia conversion at just 375°C with an NH_3 feed rate of $10\ \text{mL/min}$ — a significant improvement over conventional systems. Hydrogen permeation through the membrane provides dual benefits: product purification and kinetic acceleration of the decomposition reaction. The work establishes practical pathways for implementing ammonia as a carbon-free hydrogen carrier in energy systems.

Badakhsh et al. proposed direct Joule-heating of the Ni-Cr-Al foam catalytic reformer [25]. The thermal efficiency was demonstrated to be higher than that of previously described foam reformers. Therefore, this study shows the possibility of a Joule-heated foam reactor being used for ammonia decomposition. Despite these results, the application of metallic foam with Joule heating has not yet been well developed, as studies on this topic are insufficient and do not analyze the reactors in detail [26].

Plate small-scale ammonia reformers have garnered wide interest due to their compactness and high thermal efficiency [27–29]. These systems typically comprise metal plates with catalyst-coated surfaces, where the porous structure of the catalyst layer governs both heat transfer and reaction kinetics [25,30,31]. While various designs have been proposed — including auto-thermal configurations [32], micro-scale reactors [33,34], and electrically heated reformers — a critical knowledge gap remains regarding mass transfer limitations within the catalyst layer.

Previous studies have demonstrated that catalyst layer thickness significantly impacts reactor performance in analogous processes like steam methane reforming [35], with diffusion limitations becoming particularly pronounced in porous media [36]. However, existing research on plate-type ammonia reformers has primarily focused on thermal management and catalyst development, while systematically neglecting the investigation of diffusion constraints [37–39]. This oversight is especially critical for small-scale reformers where the high surface-to-volume ratio amplifies mass transfer effects.

Our work directly addresses this gap by investigating how catalyst layer thickness influences diffusion limitations and reaction efficiency in electrically heated plate reformers. Building upon the foundational work of Engelbrecht et al. [32] (thermal efficiency optimization) and Maleki et al. [34] (mass transfer minimization), we employ a numerical approach to quantify the non-linear relationship between catalyst thickness and ammonia conversion; identify critical thickness thresholds where diffusion becomes limiting; establish design guidelines for optimizing mass and heat transfer in compact reformers.

This systematic investigation provides essential insights for developing high-efficiency small-scale ammonia reformers, particularly for distributed hydrogen production applications where both compactness and performance are crucial.

2. Methods

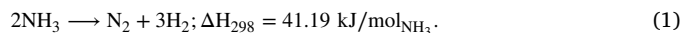
The ammonia decomposition reaction is known to be endothermic, requiring the supply of external heat to offset the heat consumption during the process according. According to chemical reaction (1), the

Table 1

Physical properties of the unmodified $\text{Ni}/\text{Al}_2\text{O}_3$ catalyst.

Parameter	Unit	Value
Additive element	–	None
BET surface area	m^2/g	140
Pore volume	cm^3/g	0.62
Pore diameter	nm	3.50
Ni surface area	m^2/g	2.05
Ni particle size	nm	29

stoichiometric equilibrium decomposition of 2 mole of ammonia yields 3 moles of hydrogen and 1 mole of nitrogen with a heat consumption of $92.38\ \text{kJ}$:



Additionally, ammonia decomposition is a catalytic reaction that takes place over a catalyst. This study focuses on analyzing a small-scale plate reformer with varying catalyst layer thicknesses. Numerical calculations were conducted under various operational conditions, and experimental results were utilized to validate the numerical model.

2.1. Geometry, mesh and catalyst

The computational domain and mesh structure for the numerical analysis of a plate mini-reformer are depicted in Fig. 1. Ammonia flows through the reformer, which is divided into different zones: inlet and outlet zones without catalyst where no heat flux is applied through the walls, and a reaction zone activated with an unmodified $\text{Ni}-\text{Al}_2\text{O}_3$ catalyst on the walls, with heat supplied through the walls. The inlet and outlet zones are added for flow development and vortex dissipation. It should be noted that, in general, for the real small-scale reformer those zones are added to get stable and developed flow. Various thicknesses of catalyst layers were investigated, with the catalyst layer represented as a porous medium. The lengths of the inlet and outlet zones are $20\ \text{mm}$ each, while the reaction zone spans $100\ \text{mm}$. The thickness of the catalyst layers ranges from 100 to $2000\ \mu\text{m}$.

The mesh structure shown in Fig. 1b is structured and designed to maintain a y^+ parameter within the near-wall region of 1. The total number of mesh elements was determined through a mesh independence study, with the degree of ammonia decomposition chosen as the control parameter. The results of the mesh independence study are illustrated in Fig. 2. To save a computational time for 2D computational domain, the total number of mesh element was set about 20,000. The number of elements slightly vary for various catalyst layer thickness.

The efficiency of ammonia decomposition depends on operational (temperature, pressure, residence time) and design (catalyst, type of reformer) parameters. There are many different catalysts for the ammonia decomposition reaction. They differ in the active components, supports and promoters. Lucentini et al. published a review on the catalyst for ammonia decomposition [14]. Unmodified $\text{Ni}-\text{Al}_2\text{O}_3$ catalyst shows a moderate efficiency with low cost; thus, these catalysts are widely used in the industry and well studied.

In this paper, an alumina-supported nickel-based catalyst ($\text{Ni}/\text{Al}_2\text{O}_3$) is chosen as catalyst in the reformer from Fig. 1. Physical properties for the unmodified $\text{Ni}/\text{Al}_2\text{O}_3$ catalyst are provided in Table 1. The physical properties of the catalyst is fully repeating those for the catalyst from the paper by Ocura et al. [40]. The catalyst powder was pelletized and grained to 10–18 mesh. The results obtained via our numerical model were validated against the experimental data from this paper [40].

It should be noted that the numerical simulation for the verification model was carried out for a tubular reformer filled with $\text{Ni}-\text{Al}_2\text{O}_3$ catalysts under conditions fully replicated from the papers by Okura et al. [40,41]. The reformer is divided into three zones: the inlet zone, the porous zone, and the outlet zone. The inlet and outlet zones are added for flow development and vortex dissipation. The porous

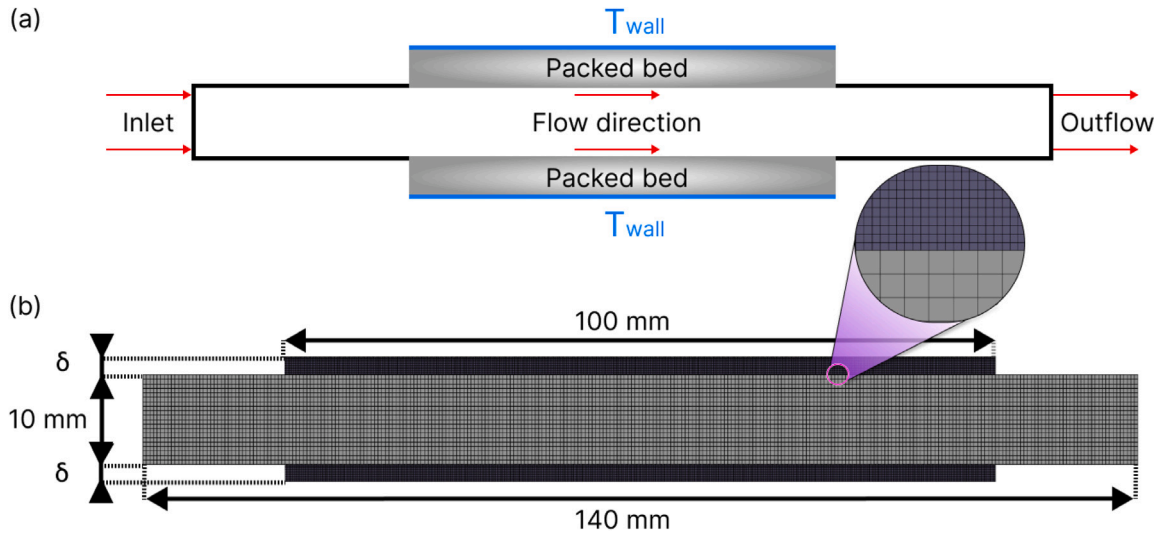


Fig. 1. Computational domain (a) and mesh (b): δ — catalyst layer thickness (scale for catalyst layer for the figure (a) is increased; the actual thickness of it is up to 2000 μm).

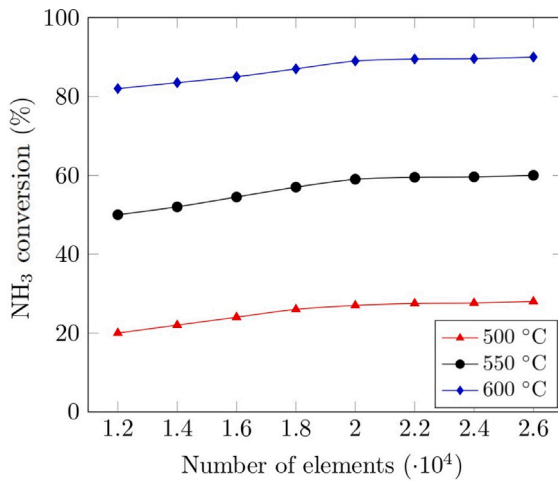


Fig. 2. Mesh independence study (reference parameter is ammonia decomposition).

zone is set as a pseudo-homogeneous packed bed filled with catalyst particles. The wall temperature is constant along the reaction zone to achieve close-to-isothermal regime. The computational domain is radially symmetric; therefore, a two-dimensional axisymmetric method was applied.

2.2. Governing equations

In the numerical model, several key processes are considered, each governed by specific equations: flow dynamics; heat transfer; flow within a porous catalyst medium; mass and heat diffusion; chemical kinetics [42,43].

The numerical model incorporates conservation equations for mass, momentum, and energy, along with equations describing the chemical reactions involved in the ammonia decomposition process and transport phenomena within the porous medium. These equations were solved using the Ansys Fluent software package.

To facilitate the computational analysis, certain assumptions were made in the development of this numerical model:

- ammonia decomposition process is steady-state process;

- no the flux of energy due to a mass concentration gradient (no Dufour effects) [44];
- work by viscous forces and by pressure is not done;
- any other surface reactions are absent;
- catalytic layer porosity is isotropic.

The mass and momentum conservation equations are as follows, respectively:

$$\nabla \cdot (\rho \vec{v}) = 0; \quad \nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\bar{\tau}) + \rho \vec{g} + \vec{F} \quad (2)$$

where $\vec{v}$ is velocity; p is the static pressure; $\rho \vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively.

The stress tensor $\bar{\tau}$ can be written as follows:

$$\bar{\tau} = \mu \left[(\nabla \vec{v} + \vec{v}^T) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \quad (3)$$

where μ is the molecular viscosity; I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

Catalytic porous media are modeled by adding a momentum source term to the momentum equation (2). The last term in Eq. (2), $\vec{F}$, incorporates additional model-dependent source terms, including the porous media contribution. In this work, we employ the semi-empirical Ergun's correlation [45,46] to describe transport phenomena in the porous catalyst. This approach is adopted due to its demonstrated validity across a wide range of Reynolds numbers, making it particularly suitable for our system.

$$S_i = - \left(\frac{150 \mu (1 - \epsilon_{ip})^2}{D_p^2 \epsilon_{ip}^3} v_i + \frac{D_p^2}{150} \frac{\epsilon_{ip}^3}{(1 - \epsilon_{ip})^2} \frac{1}{2} \rho |v| v_i \right) \quad (4)$$

where μ is the viscosity; D_p is the mean diameter of pore of the catalyst; ϵ_{ip} is the catalyst layer porosity.

The equation for energy conservation in the reaction zone has following form:

$$\nabla \cdot \vec{u}(p + \rho \cdot h_f) = \nabla \cdot \left[k_{eff} \cdot \nabla T - \left(\sum h_j \cdot \vec{J}_j \right) + \bar{\tau} \cdot \vec{v} \right] + S_f^h \quad (5)$$

where T is the temperature; ρ is the density; h_f is enthalpy of reaction mixture; $\vec{J}_j$ is the diffusion flux of j component; k_{eff} is the conductivity; h_j is the enthalpy of j component; S_f^h is the fluid enthalpy source term (include the source of energy due to chemical reaction).

The last term of Eq. (5) contains endothermic reaction heat sink (chemical energy absorption [47]) that can be defined in general form as following:

$$- \sum_{i=1}^N r_i \Delta H_{r,i} \quad (6)$$

where r_i is reaction rate of the i th reaction; $\Delta H_{r,i}$ is enthalpy change of the i th reaction (for the case of ammonia decomposition reaction $N=1$).

The effective thermal conductivity in the porous medium is computed as the volume average of the fluid conductivity and the solid conductivity:

$$k_{eff} = \epsilon k_f + (1 - \epsilon) k_s \quad (7)$$

where k_f is fluid phase thermal conductivity (including the turbulent contribution, k_t); k_s is solid medium thermal conductivity.

The energy conservation equation for the wall of catalyst zone and inlet/outlet zones is defined as follows, respectively:

$$\nabla \cdot (k_w \cdot \nabla T) = q_w; \quad \nabla \cdot (k_w \cdot \nabla T) = 0. \quad (8)$$

where k_w is the thermal conductivity of the metal wall; q_w is heat through the metal wall.

The transport processes of j th component of gas mixture can be described by following equation:

$$\nabla \cdot (\rho \cdot \vec{v} \cdot Y_j) = -\nabla \cdot \vec{J}_j + R_j \quad (9)$$

where R_j is the reaction rate of formation or consumption of j th component; Y_j is the mass fraction of j th component of gas mixture; $\vec{J}_j$ is diffusion flux of j th component.

The diffusion processes of j th component can be expressed as:

$$\vec{J}_j = - \left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t} \nabla Y_j - D_{T,j} \frac{\nabla T}{T} \right) \quad (10)$$

where $D_{j,m}$ and $D_{T,j}$ are the mass diffusion coefficient and thermal diffusion coefficient for j th species, respectively; Sc_t and μ_t are the turbulent Schmidt number and turbulent viscosity, respectively.

The effective diffusion of j th species within catalyst layer is defined based on Bosanquet approximation [48,49]:

$$D_i^{eff} = \left(\frac{1}{D_{m,i}^{eff}} + \frac{1}{D_{i,k}^{eff}} \right)^{-1} \quad (11)$$

where $D_{m,i}^{eff}$ is effective ordinary diffusivity; $D_{i,k}^{eff}$ is effective Knudsen diffusivity.

The effective ordinary diffusivity $D_{m,i}^{eff}$ is defined from:

$$D_{m,i}^{eff} = \frac{\epsilon}{\tau} \frac{1 - x_i}{\sum \frac{x_i}{D_{ij}}} \quad (12)$$

where ϵ is porosity; τ is tortuosity; x_i is molar fraction of i th species; D_{ij} is binary diffusivity for $i - j$ th pair.

The binary diffusivity D_{ij} is calculated from [50]:

$$D_{ij} = \frac{0.0143T^{1.75}}{p_i M^{0.5} \left(V_i^{1/3} + V_j^{1/3} \right)^2} \quad (13)$$

where $V_{i,j}$ — diffusion volume for i, j th species from [51]

The Knudsen diffusion coefficient is written as:

$$D_{i,k}^{eff} = \frac{\epsilon}{\tau} \frac{d_p}{3} \sqrt{\frac{8RT}{\pi \cdot M_i/2}} \quad (14)$$

where d_p is mean pore diameter.

To apply the above equations for diffusion in the numerical mode, a user-defined function (UDF) for diffusivity was written and compiled.

2.2.1. Reaction kinetics

Ammonia decomposition is an endothermic reaction (1) and its reaction rate is determined based on the kinetic model by Okura et al. [40]. The Okura et al. mechanism provides a simplified rate equation for ammonia decomposition based on several elementary steps. It relies on a pre-exponential factor k_0 , activation energy E_a and reaction order parameters α and γ :

$$R_{NH_3} = k P_{NH_3}^\alpha P_{H_2}^\gamma \quad (15)$$

$$k = k_0 e^{-E_a/RT} \quad (16)$$

where k is the reaction rate constant; k_0 is the pre-exponential factor, E_a the activation energy; P_{NH_3, H_2} is the partial pressure; R is universal gas constant; T is temperature; α and γ are fitted parameter in the Okura et al. kinetic model. All constants in Eq. (16) are taken from Okura et al. study [40].

The reaction rate constant of reaction Eq. (15) is written as:

$$k = 1.036 \cdot 10^6 \exp \left(\frac{-97800}{8.314T} \right) \quad (17)$$

The reaction order coefficients for Ni/Al₂O₃ catalyst α and γ are written as:

$$\alpha = 0.48; \quad \gamma = -0.97 \quad (18)$$

2.2.2. Boundary conditions and solution methods

The specific heat and density of the gas mixture in the reaction zone are determined using the mixing-law and incompressible-ideal-gas law equations, respectively. This calculation is feasible due to the moderate high temperature of the reaction mixture and the relatively low pressure in the reaction space (less than 10 bar). The appropriate models are chosen within the Fluent solver software. A pressure-based solver is employed, leading to the sequential solution of governing equations. Convergence criteria are set at 10^{-3} for momentum, continuity, and species, and less than 10^{-6} for energy. Hybrid initialization is utilized for the calculations. To enhance accuracy, the Second-Order Upwind approximation method is applied to the equations.

Moreover, the boundary conditions for the computational domain of the reformer shown in Fig. 1 are determined as follows:

- reformer inlet:

$$x = 0; \quad T_g = T_{g(in)}; \quad C_i = C_{i(in)}. \quad (19)$$

- reformer outlet:

$$x = L; \quad \frac{\partial C_i}{\partial x} = 0. \quad (20)$$

- reformer center:

$$y = 0; \quad \frac{\partial C_i}{\partial y} = 0; \quad \frac{\partial T_g}{\partial y} = 0. \quad (21)$$

where C_i is mole concentration of i element.

One of the main goals of this paper is to assess the intra-layer diffusion limitations. Therefore, it is expected that the transport phenomena will be primarily driven by diffusion, while conversion will be negligible due to the small pore size of the catalyst [36]. Thus, variations in inlet velocity will have a negligible effect on the transport phenomena inside the catalyst layer. Consequently, an inlet velocity was fixed for all cases related to the plate reformer. The inlet velocity for a set of numerical calculations for the plate reformer was set at 0.126 m/s, with the corresponding mass flow rate for the 2D computational domain being $3.2 \cdot 10^{-4}$ kg/s. The inlet velocity and the corresponding mole flow rate for the series of calculations for verification were set to achieve 6000 ml g⁻¹ h⁻¹ to fully replicate the experiments conducted by Okura and co-workers [40].

2.2.3. Closure relations

In this paper, Reynolds is variable and defined as follows:

$$Re = \frac{\rho v L}{\mu} \quad (22)$$

where ρ is density of fluid; v is velocity of fluid (the superficial velocity i.e. the velocity that the fluid would have through the plate reformer when porous catalyst considered as a solid); L is characteristic dimension (the distance of between the surfaces of the channel (the height of the channel)); μ is dynamic viscosity of fluid.

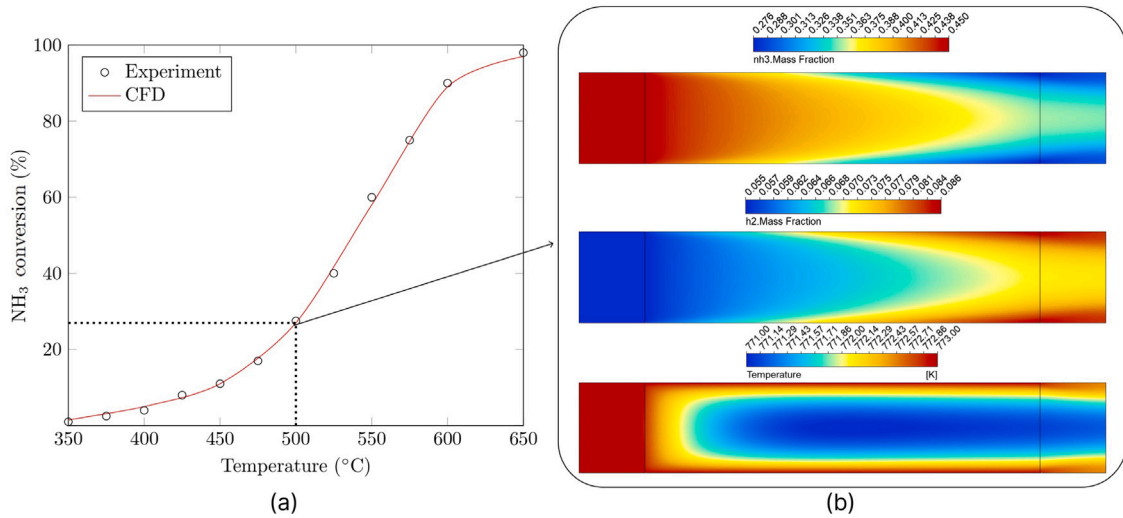


Fig. 3. The comparison of the experimental data [41] (hollow points) and the results obtained via numerical simulation (solid line).

The degree of completion of the reaction (1) (NH_3 conversion) is determined by the ammonia conversion:

$$X_{\text{NH}_3} (\%) = \frac{[\text{NH}_3]_{\text{in}} - [\text{NH}_3]_{\text{out}}}{[\text{NH}_3]_{\text{in}}} \times 100\% \quad (23)$$

where $[\text{NH}_3]_{\text{in}}$ is the molar flow rate (or number of moles) of ammonia at the reformer inlet; $[\text{NH}_3]_{\text{out}}$ is the molar flow rate (or number of moles) of ammonia at the reformer outlet.

The model by Okura et al. can be used for the intrinsic kinetics when there are no diffusion limitations within the catalyst layer. To derive the kinetic model, Okura et al. used catalyst powder, ensuring that a single catalyst particle had no diffusion limitations. However, increasing the catalyst layer thickness (or the diameter of catalyst particles) leads to greater diffusion limitations. To account for this effect in thick catalyst layers, an effectiveness factor can be introduced as follows:

$$\eta = \frac{R_{\text{real},j}}{R_{\text{intr},j}} \quad (24)$$

where $R_{\text{real},j}$ is overall reaction rate; $R_{\text{intr},j}$ is reaction rate without diffusion limitations (intrinsic kinetics).

2.3. Validation

The results obtained from numerical simulations, solving the governing equation, were compared with experimental data [41]. The additional numerical simulations aimed to validate the CFD model and used the same parameters as the experiments by Okura et al. The catalyst powder was pelletized and grained to 10–18 mesh. The mass of catalyst is 300 mg according to the second paper for the same experiments reported by Okura et al. [41]. The inlet composition of the reaction mixture consisted of NH_3 (30%), H_2 (30%), N_2 (30%), and Ar balance. The numerical simulation was performed for residence time of $6000 \text{ ml g}^{-1} \text{ h}^{-1}$ according to Okura et al. study [41].

The numerical calculations were performed at different temperatures, and the degree of NH_3 decomposition, determined using Eq. (1), was computed. Fig. 3a presents a comparison between the experimental data (hollow points) and the results obtained from numerical simulations (solid line) for a specific residence time and catalyst type. In Fig. 3a, there is a strong correlation between the experimental and numerical data, with an average deviation of less than 5%. Additionally, Fig. 3b illustrates the contours of NH_3 mass fraction, H_2 mass fraction, and temperature. These contours have physical significance and align with the contours obtained in the authors' previous papers on steam methane reforming [36].

The results presented in Fig. 3 demonstrate that the numerical model can be effectively used to calculate heat and mass transfer in other types of reformers. Therefore, the numerical model has been applied to the computational domain presented in Fig. 1.

3. Results and discussion

3.1. Contours

One of the key advantages of numerical simulations is the ability to generate detailed contours of various parameters. In Fig. 4, NH_3 mass fraction contours are depicted for a temperature of 550°C and $\text{Re} = 18.6$, considering different catalyst layer thicknesses: 100 μm , 1000 μm , and 2000 μm . The figure clearly illustrates that the thickness of the catalyst layer plays a significant role in NH_3 decomposition.

The results show that the lowest ammonia decomposition occurs with a catalyst layer thickness of 100 μm . In contrast, for catalyst layer thicknesses of 1000 μm and 2000 μm , the NH_3 mass fraction contours exhibit a similar structure. Analysis of the contours presented in Fig. 4 suggests a preliminary conclusion that the thickness of the catalyst layer influences NH_3 decomposition. However, beyond a certain threshold, increasing the thickness does not significantly alter NH_3 decomposition.

Further discussion on the impact of catalyst layer thickness on NH_3 decomposition is provided in the subsequent section below, delving into the nuances and implications of these findings.

The ammonia decomposition reaction is known to be endothermic, resulting in heat consumption within the catalyst layer due to the chemical reaction. This heat absorption causes temperature variations along the catalyst layer and flow area, as illustrated by the temperature contours depicted in Fig. 5.

Fig. 5 highlights the influence of catalyst layer thickness on temperature distribution within the plate reformer. Notably, for a thickness of 100 μm , the heat consumption stemming from the endothermic ammonia decomposition reaction is lower compared to thicknesses of 1000 μm and 2000 μm . Consequently, minimal temperature discrepancies are observed in the plate reformer with a 100 μm catalyst layer. On the contrary, the most significant temperature variation is seen with a 2000 μm catalyst layer. The main reason for the small temperature difference is that heat is transferred from the hot wall via thermal conductivity (Eq. (7)); therefore, the μ -scale catalyst layers can be considered as thermally thin. The temperature variations from Fig. 5 can be attributed to the impact of intra-layer diffusion processes on heat and mass transfer, given that the ammonia decomposition reaction

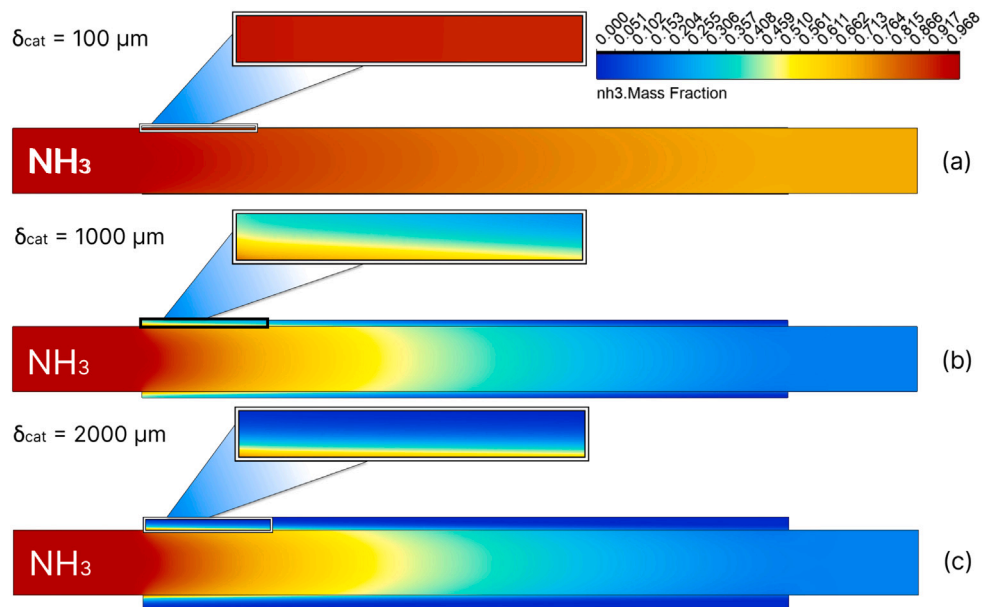


Fig. 4. The NH_3 mass fraction contours at $T = 550^\circ\text{C}$ and $\text{Re} = 18.6$ for various catalyst layer thickness: (a) $100 \mu\text{m}$; (b) $1000 \mu\text{m}$; (c) $2000 \mu\text{m}$.

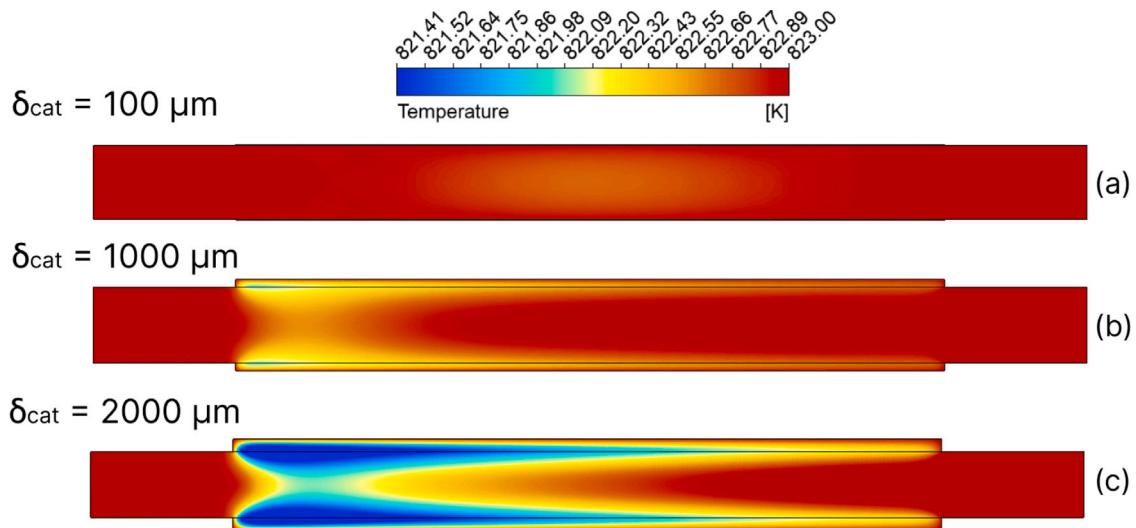


Fig. 5. The temperature contours at $T = 550^\circ\text{C}$, $\text{Re} = 18.6$ for various catalyst layer thickness: (a) $100 \mu\text{m}$; (b) $1000 \mu\text{m}$; (c) $2000 \mu\text{m}$.

occurs within the porous catalyst layer. From the side of reaction mixture, the transport phenomena inside a catalyst layer are primarily driven by diffusion (e.g., binary and Knudsen), while in the flow region, conversion is dominant. Thus, a larger catalyst layer results in greater diffusion limitations and higher temperature variation is observed. In addition, in porous media, heat and mass transfer predominantly rely on intra-layer diffusion mechanisms. Further analysis and discussion are warranted to explore the intricate interplay between catalyst layer thickness, heat consumption, temperature distribution, and intra-layer diffusion processes in the context of ammonia decomposition within plate reformers.

The ammonia decomposition process also influences the velocity distribution within the flow zone of a small-scale reformer. As depicted in the chemical reaction (1), the number of moles of products is twice that of the initial components under stoichiometric conditions. Consequently, an increase in the velocity of the reaction mixture within the reformer is anticipated. Fig. 6 illustrates the contours of velocity and the velocity distribution along the y -axis at the reformer outlet. This

visualization effectively elucidates the impact of ammonia decomposition on velocity within the reformer.

The findings presented in Fig. 6 offer insights into comprehending how the duration of contact between the reaction mixture and the catalyst layer influences decomposition performance. When catalyst layer thickness increase the velocity is increasing because the mole number of reaction products is higher than that of initial components. Therefore, more NH_3 reacted, more moles of reaction products produced. The reason for these observation is the following. Increasing the amount of catalyst increases the amount of active sites, thus making adsorption and reaction of ammonia more likely (as shown later in Fig. 7). The fact that higher conversion in a reaction including volume expansion leads to higher linear velocity (due to a fixed reactor volume) is also mostly trivial. Further increasing the amount of active sites shifts kinetic limitations towards mass and heat transport effects

Moreover, these results from Fig. 6 answer the question “How is catalyst layer thickness affected on velocity?” and show the interplay between reaction kinetics and fluid dynamics, highlighting how changes in velocity correlate with the advancement of the ammonia

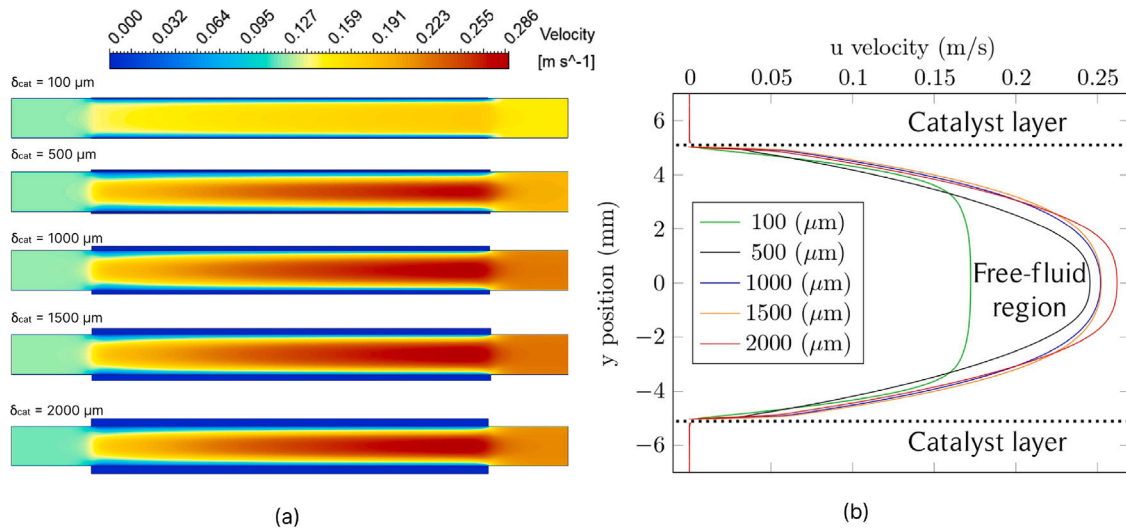


Fig. 6. The velocity contours at $T = 550\text{ }^{\circ}\text{C}$, $Re = 18.6$ (a) and the velocity distribution along the y -axis at the reformer center (b).

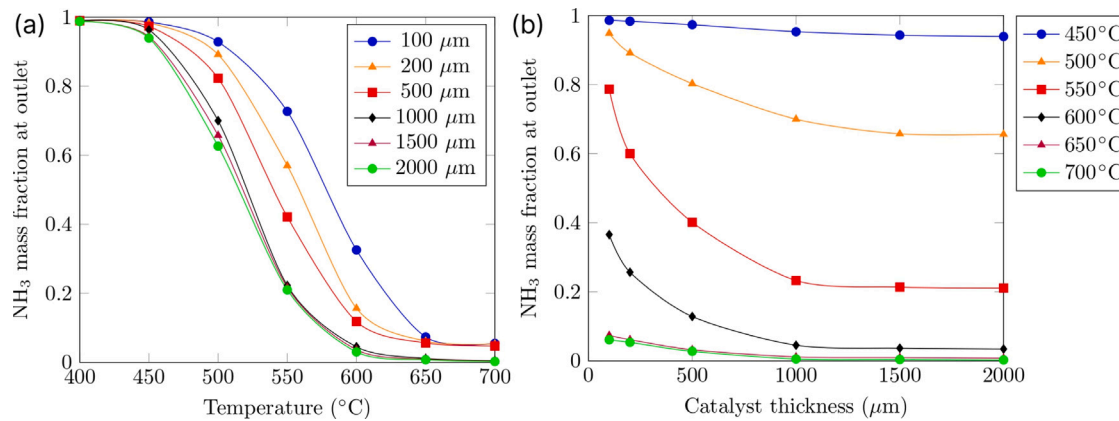


Fig. 7. NH_3 mass fraction at the reformer outlet for $Re = 18.6$ as a function of: (a) temperature for various catalyst layer thickness; (b) catalyst layer thickness for various temperatures.

decomposition process. By examining the velocity contours and distributions, researchers can discern critical patterns and trends that inform optimal design strategies for enhancing decomposition efficiency in small-scale reformers. The insights provided by Fig. 6 underscore the importance of studying not only chemical reactions but also their dynamic effects on flow behavior within catalytic reactors.

Fig. 6 shows that velocity inside the catalyst layer is negligible. Therefore, as it was mentioned in Eq. (22), the characteristic dimension of the height of channel can be used for Reynolds number calculation.

3.2. NH_3 decomposition

In order to comprehensively explore the impact of catalyst layer thickness on the ammonia decomposition process, a series of tests were conducted to investigate various operational and design parameters. Specifically, the catalyst layer thickness was varied from 100 to 2000 μm , and temperatures ranging from 400 to 700 $^{\circ}\text{C}$ were examined. Fig. 7 illustrates the relationship between the NH_3 mass fraction at the reformer outlet for $Re = 18.6$ and the catalyst layer thickness, showcasing its dependency on temperature (a) as well as catalyst layer thickness (b).

The findings presented in Fig. 7 underscore the significant influence of catalyst layer thickness on ammonia decomposition within a small-scale plate reformer operating in the temperature range of 450–650 $^{\circ}\text{C}$. It is important to note that the effect of catalyst layer thickness becomes

negligible at temperatures below 450 $^{\circ}\text{C}$ due to the near-zero reaction rate of the forward reaction. Similarly, for temperatures exceeding 650 $^{\circ}\text{C}$, the impact of catalyst layer thickness diminishes as the reaction rate increases, allowing for nearly complete ammonia decomposition within the given residence time.

Focusing on the temperature range of 450–650 $^{\circ}\text{C}$ as depicted in Fig. 7a, it becomes evident that an increase in catalyst layer thickness corresponds to a higher degree of ammonia decomposition. However, beyond a thickness of 1000 μm , there is no significant decrease in ammonia mass fraction at the reformer outlet. This observation is further supported by the data presented in Fig. 7b, which demonstrates that increasing the catalyst layer thickness beyond 1000 μm results in only a marginal reduction in ammonia mass fraction. This phenomenon can be attributed to diffusion limitations within the catalyst layer, where transport of reaction species is primarily driven by diffusion. Consequently, only a thin layer of the catalyst actively participates in the ammonia decomposition process, while deeper layers play a negligible role. Similar observations have been made in the context of steam methane reforming, where only thin surface layers of the catalyst are actively involved in the chemical process [36].

The implications of these results from Fig. 7 are significant for the design of small-scale plate reformers, suggesting that the determination of catalyst layer thickness should be based on thorough optimization investigations. Notably, for $\text{Ni-Al}_2\text{O}_3$ catalysts, increasing the catalyst layer thickness beyond 1000 μm is unlikely to yield a notable effect

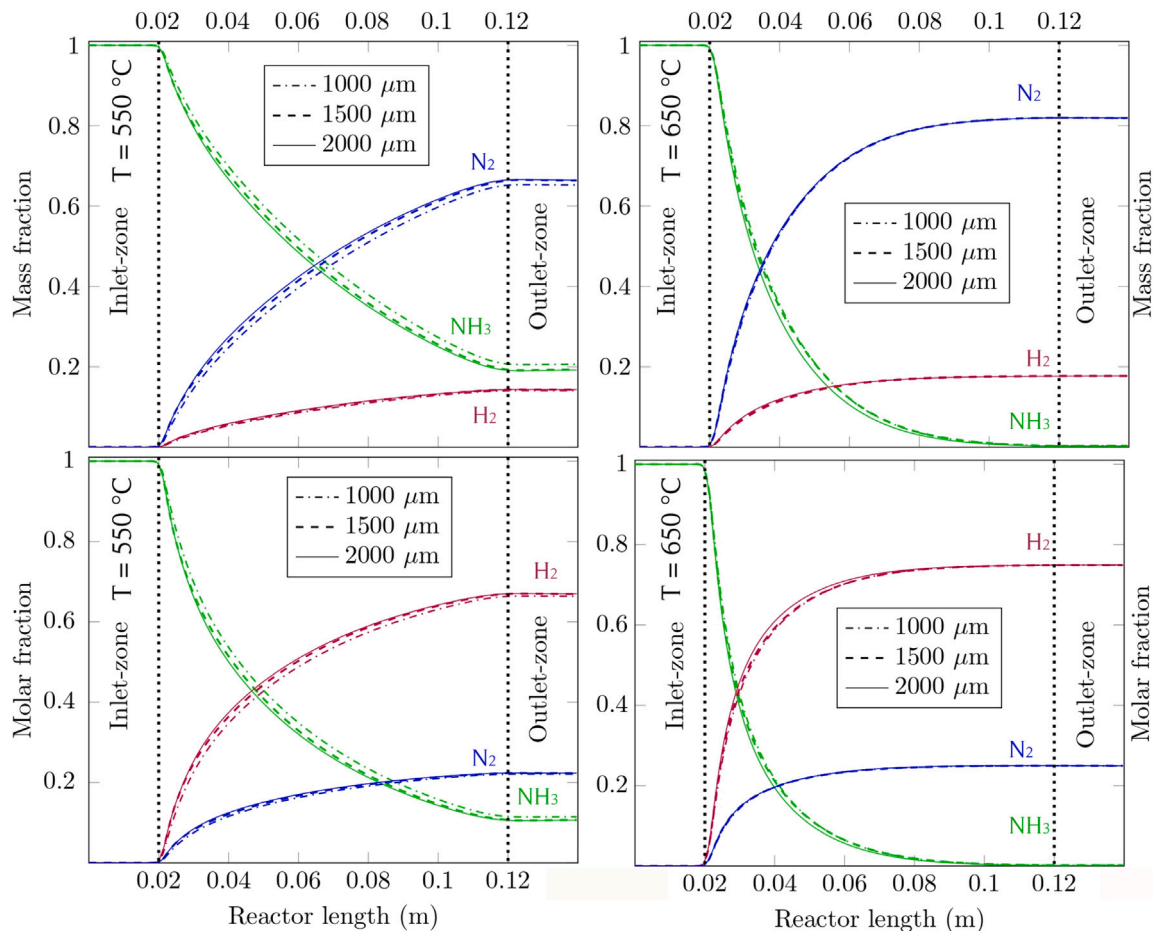


Fig. 8. Molar and mass fractions of the reaction mixture composition along the reformer length for catalyst layer thicknesses of 1000, 1500, and 2000 μm at temperatures of 550 $^{\circ}\text{C}$ and 650 $^{\circ}\text{C}$, for $\text{Re} = 18.6$, obtained from the numerical model.

on ammonia decomposition. These findings underscore the importance of carefully considering and optimizing catalyst layer thickness in small-scale plate reformer design.

In addition, the kinetic parameters for the ammonia decomposition reaction used in this model are representative for other studies of ammonia decomposition over nickel-based catalyst [52], which requires temperatures in the range of 580–650 $^{\circ}\text{C}$ to achieve high conversion levels.

In the ammonia decomposition reaction, the number of moles of reaction products is greater than the number of moles of initial ammonia, e.g. for stoichiometric reaction, two moles of ammonia produces four moles of H_2/N_2 gas. Therefore, it is important to understand the effect of pressure on ammonia decomposition. It is well-known that the small-scale plate reformers, in general, operate under the pressure below 10 bar. Table 2 shows the effect of pressure on ammonia mass fraction at the reformer outlet. The table makes it clear that in the pressure range from 1 to 10 bar the effect of pressure on ammonia decomposition is negligible and the ammonia decomposition degree differs for this pressure range for less than 1.5%. At the same time, the obtained results are fully consistent with Le Chatelier's principle, which shows that increasing pressure leads to a shift in equilibrium towards the reverse reaction.

3.3. Hydrogen-rich gas composition

One of the primary potential applications of small-scale plate reformers is in hydrogen production, as ammonia can be readily converted into hydrogen or a hydrogen-rich gas. In Fig. 8, the mass and molar fractions of species in the reaction mixture are depicted

Table 2

NH_3 mass fraction at the reformer outlet for various temperatures and catalyst layer thicknesses at different pressures of 1 and 10 bar.

Temperature, $^{\circ}\text{C}$	NH_3 mass fraction at outlet							
	400		500		600		700	
δ , μm	1bar	10bar	1bar	10bar	1bar	10bar	1bar	10bar
100	0.989	0.991	0.928	0.937	0.325	0.339	0.074	0.081
500	0.989	0.990	0.823	0.836	0.156	0.167	0.047	0.054
1000	0.987	0.989	0.699	0.712	0.118	0.126	0.005	0.011
1500	0.986	0.987	0.657	0.669	0.045	0.053	0.004	0.010
2000	0.985	0.986	0.626	0.637	0.036	0.042	0.003	0.009

for catalyst layer thicknesses of 1000 μm , 1500 μm , and 2000 μm at temperatures of 550 $^{\circ}\text{C}$ and 650 $^{\circ}\text{C}$ along the length of the reformer. Fig. 8 clearly illustrates that the mass and molar fractions for catalyst layer thicknesses of 1000 μm , 1500 μm , and 2000 μm are nearly identical. This observation validates the earlier decision that increasing the catalyst layer thickness beyond 1000 μm does not significantly affect decomposition performance.

Furthermore, Fig. 8 demonstrates that at $T = 650$ $^{\circ}\text{C}$, the reaction mixture achieves equilibrium conditions more rapidly than at $T = 550$ $^{\circ}\text{C}$. Additionally, it is evident from Fig. 8 that at $T = 650$ $^{\circ}\text{C}$, the molar fraction of hydrogen-rich gas at the reformer outlet approaches close to stoichiometric proportions, with 75% hydrogen and 25% nitrogen.

Significant variations are evident when comparing catalyst layer thicknesses of 100, 200, and 500 μm . As illustrated in Fig. 9, the catalyst layer thickness emerges as a pivotal factor influencing the distribution of mass and mole fractions within a reformer. At a temperature of

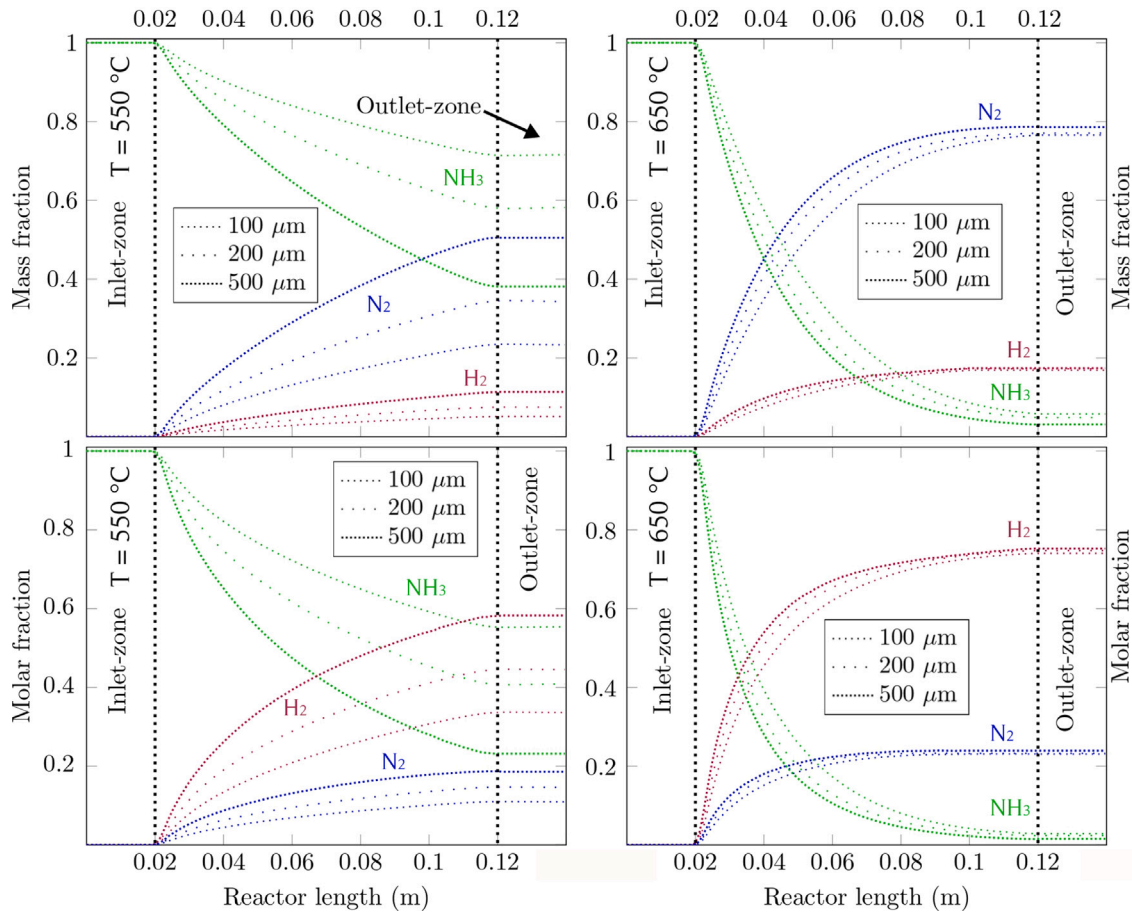


Fig. 9. Molar and mass fractions of reaction mixture composition along reformer length for the catalyst layer thickness of 100, 200, and 500 μm at temperature of 550 $^{\circ}\text{C}$ and 650 $^{\circ}\text{C}$ for $\text{Re} = 18.6$, obtained from the numerical model.

550 $^{\circ}\text{C}$, it is observed that the hydrogen molar fraction for the 500 μm thickness is approximately double that of the 100 μm thickness. This disparity diminishes with increasing temperature due to the heightened reaction rates associated with temperature increments.

Consequently, within high-temperature ranges exceeding 650 $^{\circ}\text{C}$, nearly stoichiometric product gas compositions are achieved across all examined catalyst layer thicknesses. This observation underscores that in the context of high-temperature ammonia decomposition processes, gas compositions closely approaching stoichiometry can be attained even with catalyst layer thicknesses as low as 100 μm , particularly at low Reynolds numbers.

This analysis highlights the intricate interplay between catalyst layer thickness, temperature, and resultant gas compositions in ammonia decomposition processes. The data suggests that while catalyst layer thickness exerts a significant influence, temperature variations play a crucial role in modulating the observed differences in gas compositions.

Fig. 10 presents the hydrogen-rich gas composition across the reformer length for various catalyst layer thicknesses at a temperature of $T = 550^{\circ}\text{C}$. The visual data in Fig. 10 distinctly demonstrates that increasing the catalyst layer thickness to 1000 μm at this temperature results in an increased hydrogen yield and enhanced degree of ammonia decomposition. However, a further increase in the catalyst layer thickness does not yield a corresponding increase in these parameters. This observation can be attributed to intra-particle diffusion limitations.

The transport of species within the porous catalyst layer is predominantly governed by diffusion, where the diffusion rate is notably slower than that of convection. As a result, once the catalyst layer thickness surpasses a certain threshold, any subsequent increase insignificantly impacts the performance of ammonia decomposition. This limitation

underscores the critical role of intra-particle diffusion in governing the effectiveness of ammonia decomposition within the catalyst layer.

3.4. Effectiveness factor

To determine the diffusion limitations within the catalyst layer for ammonia decomposition, the effectiveness factor was calculated according to Eq. (24). The effect of catalyst layer thickness and temperature on the effectiveness factor is presented in Fig. 11. Fig. 11a shows that an increase in catalyst layer thickness significantly reduces the effectiveness factor and, as a result, increases the diffusion limitations within the catalyst layer. For example, the diffusion limitations for a catalyst layer thickness of 2000 μm is approximately five times higher than that for a catalyst layer thickness of 100 μm . Fig. 11a also indicates that temperature has a minor effect on the diffusion limitations. The quantitative comparison of the effect of temperature on the effectiveness factor is illustrated in Fig. 11b, which clearly shows that an increase in temperature leads to an insignificant increase in diffusion limitations within the catalyst layer. One of the main reasons for this observation can be found in the diffusion mechanism for porous media. According to the Bosanquet approximation, the effective diffusion depends on binary diffusion and Knudsen diffusion. However, for porous media, the diffusion limitations via the Knudsen mechanism is significantly greater than that for binary diffusion, primarily due to the linear dependence of the Knudsen diffusion coefficient on pore size. Therefore, the catalyst layer thickness has greater effect on the diffusion limitations than temperature.

The dependencies in Fig. 11, obtained from numerical simulations, can be expressed as functions of both the catalyst layer thickness and temperature.

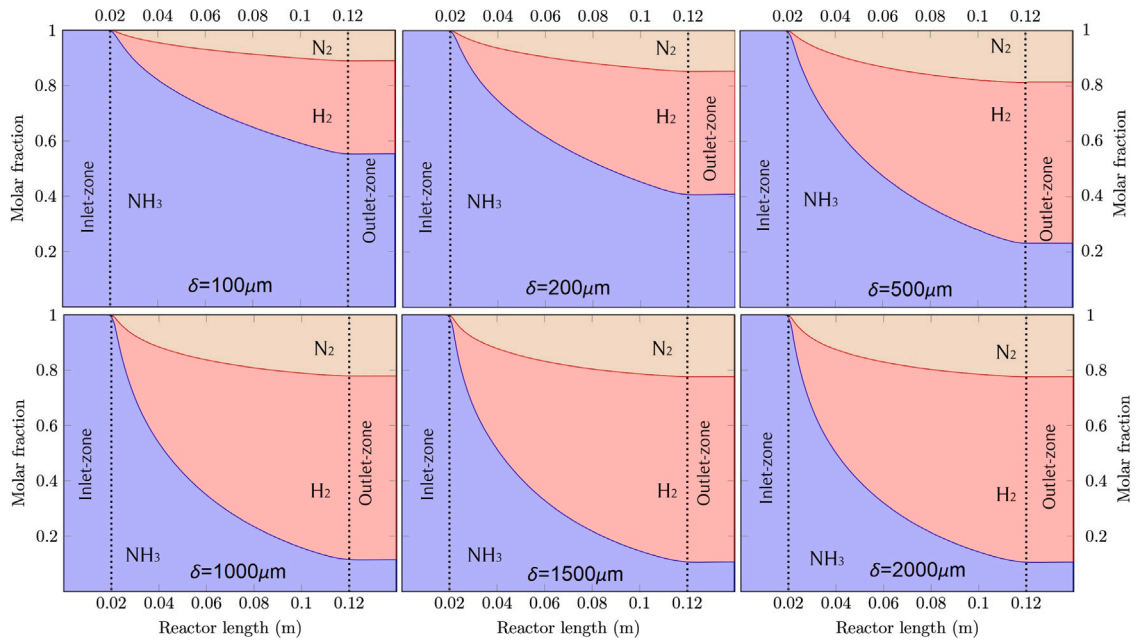


Fig. 10. The hydrogen-rich gas composition along the reformer length for various catalyst layer thicknesses at $T = 550^\circ\text{C}$ and $Re = 18.6$, obtained from the numerical model.

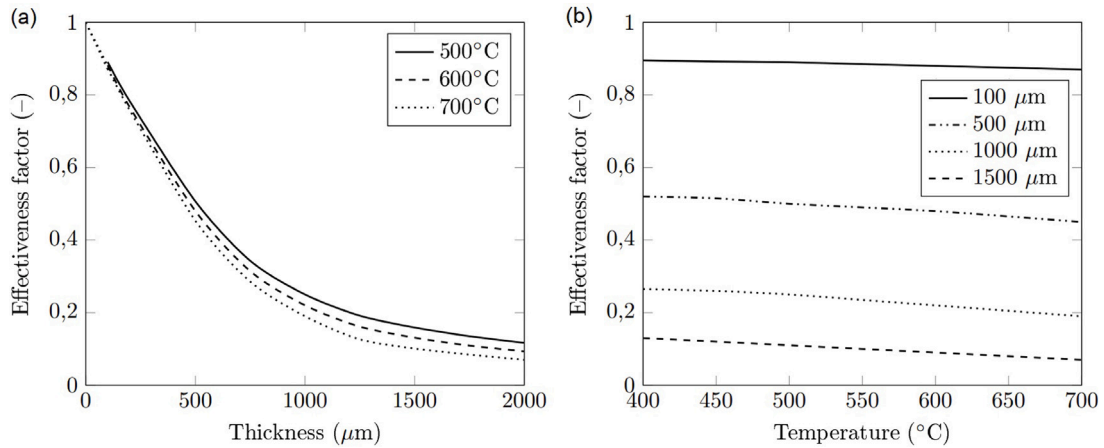


Fig. 11. Effectiveness factor as a function of catalyst layer thickness (a) and temperature (b).

Based on our analysis, the effectiveness factor can be calculated as a function of catalyst layer thickness and temperature using the following equation:

$$\eta = f_{\delta}(\delta) \cdot f_T\left(\frac{T}{1000}\right), \quad (25)$$

where $f_{\delta}(\delta)$ is the thickness-dependent component of the effectiveness factor; $f_T\left(\frac{T}{1000}\right)$ is the temperature-dependent component of the effectiveness factor.

Taking into account the obtained results, the functions f_{δ} and f_T from Eq. (25) were derived using regression analysis. For the thickness-dependent function $f_{\delta}(\delta)$ and temperature-dependent functions f_T the following correlations were established:

$$f_{\delta}(\delta) = 0.7 \cdot e^{-0.002\delta} + 0.2; \quad (26)$$

$$f_T\left(\frac{T}{1000}\right) = -1.2564 \cdot \left(\frac{T}{1000}\right)^2 + 1.9213 \cdot \left(\frac{T}{1000}\right) + 0.3754. \quad (27)$$

where δ is the catalyst layer thickness in μm .

4. Conclusion

In this paper, the ammonia decomposition process in a small-scale plate reformer was investigated using a 2D planar CFD-based model approach coupled with chemical kinetics, porous media, and diffusion models. The results revealed the significant influence of catalyst layer thickness, particularly up to $1000 \mu\text{m}$, on the ammonia decomposition process within the temperature range of 450 – 650°C . The impact of catalyst layer thickness becomes smaller at temperatures below 450°C due to the near-zero reaction rate of the forward reaction. Similarly, at temperatures exceeding 650°C , the effect of catalyst layer thickness decreases as the reaction rate increases, enabling nearly complete ammonia decomposition within the given residence time.

The intra-layer diffusion limitations was defined via an effectiveness factor and results showed the significant effect of catalyst layer thickness on it, e.g. the diffusion limitations for a catalyst layer thickness of $2000 \mu\text{m}$ is approximately five times higher than that for a catalyst layer thickness of $100 \mu\text{m}$. At the same time, the effect of temperature on the intra-layer diffusion limitations is minimal. Additionally, the results indicated that within the temperature range up to 600 – 650°C , increasing

the catalyst layer thickness beyond 1000 μm does not proportionally enhance ammonia decomposition and hydrogen yield. However, for high-temperature ammonia decomposition (above 650–700 $^{\circ}\text{C}$), where the reaction rate is significantly higher, close-to-stoichiometric product gas composition can be achieved with a thinner catalyst layer.

Further studies are needed to evaluate process performance and determine intra-catalyst diffusion limitations in industrial-scale reformers. In addition, it is important to analyze other catalyst types and provide techno-economic optimization of the catalyst layer thickness for small-scale plate reformers because as it is shown in the presented paper the linear increase in catalyst layer thickness is non-linear increasing the ammonia decomposition.

CRedit authorship contribution statement

Viacheslav Papkov: Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Writing – original draft. **Leonid Tartakovsky:** Writing – review & editing, Supervision, Resources, Conceptualization, Investigation. **Dmitry Pashchenko:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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